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GITHUB LINK

Plant Health Classification

Predictive Classification

Scenario: Campus grounds personnel want to classify plant condition from visible symptoms, soil moisture, and growth indicators before recommending a response.

System question: Based on the recorded inputs, which category should the system assign for plant health classification?

Implementation scope	Compare the primary classifier with one alternative
Primary algorithm	Decision Tree Classifier
Compare with	Random Forest
System input	Leaf color, spots, wilting, soil moisture, growth rate
System output	Plant condition and suggested response
Python method	from sklearn.tree import DecisionTreeClassifier

Part 1 Understand the Dataset

Class activity: Identify reference-only fields, analytical inputs, any required target label, the generated output, missing information, and one privacy, fairness, safety, or accessibility concern.

Part 2 Prepare and Check the Dataset

Minimum requirement: Create approximately 30 to 100 records using four to eight relevant input features. Check missing values, duplicates, invalid categories, and target balance. Exclude identifiers and encode categorical inputs consistently.

Part 3 Build the KNIME Workflow

Suggested workflow: CSV Reader -> Column Filter -> Missing Value -> One to Many -> Partitioning -> Learners -> Predictors -> Scorer

Evaluation: accuracy, precision, recall, F1-score, confusion matrix

Part 4 Recreate the Process in Python

Required process: Use from sklearn.tree import DecisionTreeClassifier. Load the CSV with pandas, prepare X and y, split or cross-validate, train the listed classifiers, compare metrics, and save the selected pipeline with joblib.

Part 5 Create the Streamlit Interface

Interface requirement: Collect or upload these inputs: Leaf color, spots, wilting, soil moisture, growth rate. Validate incomplete or invalid values, run the same Python process, and display plant condition and suggested response with a brief explanation.

Part 6 Test Explain and Reflect

Testing: Test normal, missing, boundary, and uncertain cases. Compare the primary and comparison algorithms using one results table or appropriate chart.

Critical-thinking questions:

1. Which input features should have the strongest influence on the assigned category?
    - Leaf color abnormalities, visible spots, and wilting should carry the strongest predictive influence because they are direct, acute visual indicators of physical distress, disease, or pest damage. While soil moisture and growth rate provide essential contextual background for chronic stress, sudden changes in leaf appearance and structural integrity typically manifest the fastest and most severe signs of plant decline.
2. What should the system do when the available evidence is incomplete or contradictory?
    - When evidence is incomplete or contradictory, the system should flag the record as uncertain or low-confidence rather than forcing a definitive classification. It should prompt the user or campus personnel to provide the missing fields and recommend a manual, physical inspection by an expert before executing any aggressive treatment, ensuring safety against misdiagnosis.
3. What could happen if the system produces an incorrect result for plant health classification?
    - An incorrect classification could lead to inappropriate or harmful interventions—such as overwatering an already root-rotted plant, applying the wrong chemical treatment for a misidentified disease, or ignoring a contagious pathogen. This could result in the death of valuable campus flora, wasted labor and material resources, and potentially the rapid spread of disease to neighboring plants across the grounds.

DATA DICTIONARY:

Column	Description	Data type	Valid values or range	Role
leaf_color	**Observed color of the plant leaves.**	**Categorical**	**Light Green, Pale Green, Green, Dark Green, Yellow, Brown**	**Input feature**
spots	**Amount of visible spots on the leaves.**	**Categorical**	**None, Few, Many**	**Input feature**
wilting	**Level of visible wilting.**	**Categorical**	**None, Mild, Severe**	**Input feature**
soil_moisture	**Soil moisture measurement.**	**Numeric**	**0 to 100 percent**	**Input feature**

growth_rate	**Recent rate of plant growth.**	**Categorical**	**Slow, Normal, Fast**	**Input feature**
plant_condition	**Overall health category assigned to the plant.**	**Categorical**	**Healthy, Mild Stress, Severe Stress, Diseased**	**Target label**
suggested_response	**Recommended action based on the assigned plant condition.**	**Text**	**Care or inspection recommendation**	**Output only; exclude from training**

Plant Health Classifier

Enter the plant inspection information to classify the plant's current condition.

This prediction supports campus grounds monitoring. It is not a final horticultural diagnosis.

Plant name or location (optional)

Example: Rose bed A

Leaf color

Light Green

Leaf spots

Many

Wilting

None

Soil moisture (%)

15.00

Growth rate

Slow

Classify Plant Health

Classification Result

Prediction: Mild Stress

Recommended action: Check watering, sunlight, and soil drainage.

Model confidence

57%

Human review recommended before action is taken.

› View submitted information

Predictions should be reviewed by campus grounds personnel before treatment or removal decisions.

Leaf color

Light Green

Light Green

Pale Green

Green

Dark Green

Yellow

Brown

Leaf spots

Many

None

Few

Many

Wilting

None

None

Mild

Severe

Growth rate

Slow

Slow

Normal

Fast